Paper-Writing Workflow

GRBV Transcriptomics Study

A comprehensive guide for transforming 35 figures, 13 data tables, and 10 analysis scripts into a publication-ready manuscript and conference presentation.

Quick Reference

Dataset: 327 samples, 14,604 genes, 2 seasons, 4 maturity stages

Figures ready: 35 publication-quality PNGs

CSV tables: 13 supplementary data files

Draft report: GRBV_Transcriptomics_Report.md (500 lines)

PPT: GRBV_Transcriptomics_Presentation.pptx (auto-generated)

Phase 1: Organize Your Story Arc

Day 1

1.1 Define the Central Narrative

Your data supports this story:

> ***GRBV hijacks host transcription in a development- and environment-dependent manner, suppressing heat-stress defenses, inverting seasonal response kinetics, and targeting genes that co-localize with known disease-resistance and berry-quality QTLs -- with implications for diagnostics and breeding.***

1.2 Recommended Paper Structure

Section	Key Content	Figures	Tables
Abstract	Multi-dimensional RNA-Seq; 6 research Qs; inverted trajectory; biomarkers	--	--
Introduction	GRBV biology, RBD symptoms, knowledge gaps, objectives	Fig 23	--
Methods	RNA-Seq pipeline, DE analysis, pathway, biomarker, QTL, heat-stress	--	Table 1
Results 1: Global	PCA, variance hierarchy (stage > season > infection)	Fig 1	--
Results 2: Duration (Q1)	Biphasic pattern, early vs late responders	Fig 2, 8	Table S1
Results 3: Stage (Q2)	Stage-dependent response, Post-veraison/Harvest peaks	Fig 3, 9	Table S2
Results 4: Season (Q3-Q5)	KEY: Inverted trajectory, environment modulation	Fig 4-6, 10-12, 15-19	Table S3-S5
Results 5: Trajectory (Q6)	Biphasic: acute defense -> chronic metabolism	Fig 7, 13	Table S6
Results 6: Pathways	Phenylpropanoid, plant-pathogen, ABC transporters	Fig 14	Table S7
Results 7: Heat-Stress	115 heat genes, sHSP suppression, virus-heat split	Fig 34-35	Table S4-S6
Results 8: Env-Titer	Temperature-titer-DEG correlations	Fig 30-33	Table S9
Results 9: Biomarkers	Top 50 biomarkers, 3-gene panel (AUC=1.0)	Fig 20-22	Table S1
Results 10: GWAS/QTL	60% co-localization, resistance clusters, sugar QTL	Fig 24-25	Table S2
Discussion	Developmental modulation, seasonal inversion, breeding	Fig 26-29	--
Conclusions	6 key conclusions	--	--

1.3 Suggested Journal Targets

Journal	Impact	Fit	Notes
Plant Cell	~17	Excellent	Multi-dimensional systems biology, mechanistic insight
New Phytologist	~10	Excellent	Plant-pathogen interaction, environmental modulation
Plant Physiology	~8	Very Good	Comprehensive transcriptomics, heat-stress biology
Mol. Plant-Microbe Interact.	~5	Very Good	Virus-host interaction focus
Frontiers in Plant Science	~5	Good	Broad plant biology, open access

Phytopathology	~3	Good	Plant disease focus

Phase 2: Figure Selection & Refinement

Day 2-3

2.1 Main Text Figures (recommended 6-8)

Priority	Figure	File	Rationale
1	Analysis Pipeline	23_analysis_schematic.png	Orients the reader
2	Global PCA	01_global_pca_overview.png	Shows variance hierarchy
3	Seasonal Maturity DEGs	15_seasonal_maturity_DEG_comparison.png	KEY: inverted trajectory
4	Pathway Landscape	14_master_pathway_landscape.png	Systems-level view
5	Heat vs Virus	35_heat_vs_virus_disentanglement.png	Novel biology
6	Biomarker Overview	20_biomarker_overview.png	Translational impact
7	QTL Chromosome Map	24_biomarker_QTL_chromosome_map.png	Breeding relevance
8	Summary Dashboard	29_comprehensive_summary_dashboard.png	Integrative conclusion

2.2 Supplementary Figures

All other figures (02-07, 08-13, 16-19, 21-22, 25-28, 30-34) go as supplementary material.

2.3 Figure Refinement Checklist

- [ ] Resolution >= 300 DPI (current: 150 DPI -- re-render at 300)
- [ ] Font size >= 8pt in all panels
- [ ] Consistent color scheme across figures
- [ ] Panel labels (A, B, C, D) clearly visible
- [ ] Axis labels fully spelled out (no abbreviations)
- [ ] Legend positioned without overlapping data
- [ ] Color-blind friendly palette (check with Coblis simulator)

To re-render at 300 DPI: In each .py script, change 'figure.dpi': 150 to 'figure.dpi': 300.

Phase 3: Write the Manuscript

Day 3-10

3.1 Writing Order (most efficient)

1. Methods -- straightforward, sets the foundation (use Report Section 2)
2. Results -- present findings in logical order (use Report Section 3 as draft)
3. Discussion -- interpret results, compare to literature

4. Introduction -- frame the problem knowing what you found
5. Abstract -- summarize after everything else is written
6. Title -- finalize last

3.2 Key Numbers to Cite

```
Dataset:         327 samples, 14,604 genes (from 28,168 raw)
Infection groups: RB(+) infected vs RB(-) uninfected controls
Seasons:         2021 (cooler) and 2022 (hotter, +7.6C at Harvest)
Stages:          Pre-veraison, Veraison, Post-veraison, Harvest
Cohorts:         7 infection-year cohorts (2016-2022)

DE thresholds:   padj < 0.05, |log2FC| >= 1
Enrichment:      Fisher's exact, BH FDR correction

Key DEG counts:
  - 2021 Harvest: 1,192 DEGs (peak in 2021)
  - 2022 Post-veraison: 565 DEGs (peak in 2022)
  - Cross-season correlation: Pearson r = 0.33-0.69

Heat-stress:
  - 115 heat-stress genes expressed (9 categories)
  - 57 heat genes are GRBV DEGs
  - 29/36 sHSPs downregulated at 2021 Harvest in RB(+)
  - 23 heat genes with BOTH virus and environment effects
  - HSF-sHSP module correlation: r=0.625, p=7.2e-37

Biomarkers:
  - 12 universal biomarkers (robust across all conditions)
  - 3-gene panel AUC = 1.000
  - 60% of biomarkers near known QTLs (within 1 Mb)
  - 18 biomarkers in disease-resistance clusters
  - Top host biomarker: F-box/LRR protein (AUC = 0.880)

Environmental:
  - 2022 Harvest was +7.6C warmer than 2021
  - sHSP module score: +1.64 (2022 Harvest RB-) vs +0.46 (2021)
  - GRBV reduces sHSP induction by ~50% (RB+ vs RB-)
```

3.3 Manuscript Draft Sources

Section	Source File	Lines
Introduction	GRBV_Transcriptomics_Report.md	29-41
Methods	GRBV_Transcriptomics_Report.md	44-94
Results (Q1-Q6)	GRBV_Transcriptomics_Report.md	98-394
Discussion	GRBV_Transcriptomics_Report.md	396-425
Conclusions	GRBV_Transcriptomics_Report.md	435-448
Heat stress results	heat_stress_signatures.py output	(run script)

Env-titer results	environment_titer_integration.py output	(run script)

3.4 Literature to Cite

- GRBV original characterization (Al Rwahnih et al., 2013)
- GRBV genome and classification (Varsani et al., 2017)
- GRBV transcriptomics (Blanco-Ulate et al., 2017; Rumbaugh et al., 2021)
- GRBV impacts on berry metabolism (Girardello et al., 2020)
- Grapevine heat-stress HSPs (Liu et al., 2012; Wang et al., 2014)
- Grapevine QTL/GWAS resistance loci (Zyprian et al., 2016; Possamai et al., 2020)
- Biomarker RT-qPCR diagnostics in plants (various)

Phase 4: Supplementary Materials *Day 8-12*

4.1 Supplementary Tables

Table	Source CSV	Description
S1	GRBV_biomarkers_top50.csv	Complete biomarker list with scores
S2	GRBV_biomarker_QTL_crossref.csv	QTL cross-reference details
S3	universal_DEGs_across_approaches.csv	908 robust DEGs
S4	heat_stress_DEGs_all_conditions.csv	Heat-gene DE results
S5	heat_gene_temperature_correlations.csv	Temperature correlations
S6	heat_DEGs_recurrence.csv	Recurrent heat DEGs
S7	Q1_pooled_vs_stratified.csv	Analytical robustness (Q1)
S8	Q2_pooled_vs_stratified.csv	Analytical robustness (Q2)
S9	integrated_environment_titer_DEG.csv	Env-titer-DEG integration

4.2 Data Availability Statement

```
Raw RNA-Seq data have been deposited in [NCBI GEO/SRA under accession GSEXXXXX].
All analysis scripts, processed count matrices, and supplementary data files are
available at [GitHub repository URL]. The Vitis vinifera functional annotation
database (org.Vvinifera.eg.db) was obtained from Bioconductor.
```

Phase 5: Review & Polish *Day 12-15*

5.1 Internal Review Checklist

- [ ] All figure references in text match actual figure numbers
- [ ] All p-values and fold-changes verified against script outputs
- [ ] Methods section matches what scripts actually do (check thresholds)
- [ ] Supplementary table column headers explained
- [ ] No orphan references (every citation in text is in bibliography)
- [ ] Author contributions drafted
- [ ] Acknowledgments and funding sources listed
- [ ] Conflict of interest statement prepared

5.2 Common Reviewer Concerns

Concern	Pre-emptive Response
Why Welch's t-test?	Large sample (n=327) makes parametric robust; show supplementary DESeq2 comparison
Multiple testing burden?	Each question uses independent BH correction; 908 universal DEGs validated
Batch effects between seasons?	Addressed by Q5 (baseline environmental in RB- controls); no PCA batch clustering
Small sub-group sample sizes?	Minimum n=3 per group; stratified analysis confirms pooled results
Biomarker validation?	RT-qPCR as future work; 5-fold stratified cross-validation already performed

Phase 6: Generate Presentation *Automated*

Run the following command to generate the slide deck:

```
python3 generate_presentation.py
```

This produces GRBV_Transcriptomics_Presentation.pptx with all 35 figures organized into a 44-slide conference-ready deck with section dividers, key-number panels, bullet-point annotations, and speaker notes.

File Inventory

Analysis Scripts (10)

1. analysis.py -- Core DE analysis (Q1-Q6), PCA, Figs 1-7
2. pathway_enrichment.py -- GO/KEGG enrichment, Figs 8-14
3. seasonal_maturity_progression.py -- Season x Stage, Figs 15-19
4. biomarker_discovery.py -- Biomarker scoring & classification, Figs 20-22
5. gwas_qtl_crossref.py -- QTL cross-reference, Figs 24-25
6. stratified_vs_pooled_comparison.py -- Robustness check, Figs 26-29
7. environment_titer_integration.py -- Weather + titer, Figs 30-33
8. heat_stress_signatures.py -- Heat-stress signatures, Figs 34-35
9. analysis_schematic.py -- Pipeline diagram, Fig 23
10. generate_report_pdf.py -- PDF report generator

Key Output Files

- GRBV_Transcriptomics_Report.md -- Full draft report (500 lines)
- GRBV_Transcriptomics_Presentation.pptx -- 44-slide conference deck
- 35 PNG figures (01-35) -- Publication-quality plots
- 13 CSV files -- Supplementary data tables